# PREDICTION OF NOISE EXPOSURE LEVLES USING SIMULATED FLIGHT TRAJECTORIES

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## Abstract

Noise models, especially aircraft noise models, are needed for the prediction of noise levels where noise measurements are not feasible or possible. In most cases it is prohibitively expensive for airports to measure aircraft noise in their wider surroundings. Thus, as the number of noise measurement stations is limited, it is not possible to generate detailed contours of the noise exposure levels by using real measurements.

For this reason analytical noise models are commonly used to compute noise exposure contours for the vicinity of airports on the basis of recorded radar data sets or simulated flight trajectories.

This paper discusses different aircraft noise calculation models and analyses their advantages and disadvantages.


## 1 Introduction

The increased number of aircraft movements leads to the situation that aircraft noise is anticipated as environmental problem. Noise exposure is annoying and can, especially in the vicinity of intensely frequented airports, be harmful to health. Noise is an individual stress factor and depends on the tolerance level of each person's reaction to noise.

Noise in general is a common expression for undesirable sound. Sound is the physiological sensation perceived by one of the human senses, the ear. It is classified into two categories, musical sound and noise. Musical sound produces a pleasing sound of a series of similar impulses, one following the other in a regular and continuous stream. Noise, on the other hand, is a sound that lacks the just mentioned characteristics because of irregular periods. Noise can be defined as unwanted sound or sound in the wrong place at the wrong time. [1]

The investigation of aircraft noise can be done in two ways, either with calculation or with measurement. Measurement can only capture the actual state, while calculation can also show the future or a modified noise abatement measures. With noise calculation software it is possible to show large-scale noise contours of past and future situations. Two European aircraft noise model methods were implemented by the author. The results of the calculated noise values were compared to an existing software product and validated with real measurements.

The remainder of this paper is structured as follows: Section 1 gives an introduction to this paper. In section 2 the background of aircraft noise modeling is described. Section 3 explains the referred the aircraft noise models in this paper. Section 4 describes the common database used for aircraft noise calculations. Section 5 shows the results and validation and section 6 gives an outlook for future research.

## 2 Background

Already in the 70s Germany was one of the first countries in the world which developed rules on how to calculate flight noise annoyance (the first version of the so called AzB [2] was produced in 1975). The guideline included a detailed description of how data had to be collected and prepared, stepwise instructions for the calculation, and how to illustrate these results. The first version of AzB used a simple approach for the aircraft noise calculation, Closest Point of Approach (CPA).

AzB is a german acronym and stands for "Anleitung zur Berechnung von Fluglärm", which would translate to "instructions to calculate aircraft noise" is now available in a new version published in 2008 and known as AzB2008. The current modeling approach is based on a segmentation model in order to achieve higher accuracy.

In the 80s the US aviation agency FAA (Federal Aviation Administration) published its own aircraft noise calculation model based on a segmentation approach. The model's first version, SAE-AIR 1845, is now the most common and frequently used Integrated Noise Model, INM. [3]

In the mid 80s, the European Civil Aviation Conference (ECAC) and the International Civil Aviation Organization (ICAO) both developed guidance on aircraft noise modeling for civil airports. These documents described the same "best practice" methodology for the calculation of aircraft noise and focus on the algorithms to be implemented in computer programs. However, these documents did not provide the aircraft-specific noise and performance data required for the implementation of the actual noise model. [4]

ECAC has updated the guidance documents two times since their first publication. The latest version is ECAC.CEAC Doc. 29 3rd Edition published in two parts in 2005: Volume 1, the Applications Guide [4] and Volume 2, the Technical Guide [5].

Of the models investigated in this document, FLULA2 [6] is the newest model. It was developed to model aircraft noise around civil and military airports. Unlike the other models, FLULA2 uses a simulation approach.

## *Aircraft Noise Modeling Methodology*

There are several widely used approaches for the calculation of noise levels.

In principle there are two different path-integration approaches for the computation of duration dependent event levels $L_E$: Simulation based models and segmentation models:

- Simulation based models calculate the received sound levels for a sequence of points before integrating them over time.
- Segmentation models sum contributions from discrete segments of the flight path.

Each noise calculation approach has its strengths and weaknesses.

Simulation based models are more time consuming, however, they offer very high degrees of accuracy. For the realization of such models a huge amount of data is needed. E.g. the exact flight path, accurate data of the aircraft type, soil conditions (topography), etc.

Segmentation models are faster than simulation models, although not quite as exact. Most noise models use the segmentation approach because the needed data is less complex than for simulation based models.

Over the last decades it was mostly computing power that determined the usage of the available approaches: In the early 70s, when a first version of AzB [2] was developed, only very limited computing power was available and manual calculations were still common. Thus it was necessary to use simple models. With the increase of computing power segmentation models were used, although a lot of the computation had still to be done manually. E.g. for the updated AzB [7], a companion document, the "Anleitung zur Datenerfassung" AzD (Guide for data recording), was needed to perform the calculation.

The legacy noise models did not have the opportunity to access public noise databases as they are available today. In addition, the increased computing performance and the ability to use noise databases make detailed simulations of flight traffic now possible. Therefore simulation models become feasible. At the time of this writing a small number of simulation models are already in use, e.g. FLULA2 [6]. However, they still require huge databases to achieve good results. Therefore segmentation models are still more common, as the gathering of data for simulation models can be cumbersome and expensive.

### Closest Point of Approach (CDA) Models

CPA models use relatively simple algorithms and are very fast. The noise impact at the observer-point is calculated with the A-weighted maximum value of sound levels that occur during an event ($L_{Amax}$) at the Closest Point of Approach (CDA) from the aircraft (source) to the receiver (measurement station). The German noise model AzB [2], issued in 1975, used this approach.

### Integrated Models

Noise data of straight over flights with standardized distances and predefined engine power is given in the Noise-Power-Distance Tables (NPD-Tables). The complete fly-by is indicated by an integrated noise value. To adjust the impact of noise at the receiver, several corrections (such as lateral attenuation, curved flight or speed) can be used.

In collaboration with EUROCONTROL an updated database of the Noise-Power-Distance tables (NPD) called "Aircraft Noise and Performance database" (ANP) [8] is accessible.

**Segmentation Models**

Segmentation models are a significant improvement over integrated models due to the splitting of flight path into segments. With the concept of segmentation for curved flights and variable power settings especially for the starting phase, these models allow more realistic representations of flights.

The higher accuracy of the flight path calculation is achieved by dividing the flight path into segments. This allows better modeling of spiraling and reduction of fan power after the take-off phase.

Each segment has individual power settings and speed settings providing a more realistic approximation. Segmentation models use the same Noise Power Distance values as integrated models to compute noise.

Well-known models are the Integrated Noise Model INM (USA) based on the revised ECAC.CEAC DOC.29 3rd Edition [4], NORTIM (N) and DANSIM (DK).

**Simulation Models**

For simulation models it is important to model how aircraft emit sound in different directions. Therefore, the flight path is broken down into a series of discrete points, where the sound propagation is calculated for every receiver point on the flight path. For each point, the geometry from the source to the receiver is clearly defined. The concatenation of these levels produces a time history of the sound levels analogue to the real situation. However, the usage of simulation models demands more data than just the Aircraft Noise and Performance database (ANP). To use the simulation approach it is necessary to have detailed radar data or detailed simulated data of the flight path with aircraft positions for every few seconds. Therefore, segmentation is not needed anymore because the segment is implicitly given as the distance between the last aircraft position and the current aircraft position. This allows a more accurate calculation of noise values for circular arcs, because of the high quantity of flight path positions. The border between segmentation and simulation models is blurred if the flight path segmentation length of the segmentation model gets smaller.

The Swiss model FLULA2 [6] (developed by EMPA) is using the simulation model.

**Aircraft Grouping and Substitution**

It is possible to create air traffic scenarios with an accuracy down to the exact aircraft types, however, it is often more viable to group similar aircraft types and substitute them with aircraft categories. According to ECAC Doc.29 [4] further reasons for creating various categories may be a lack of data for individual aircraft models or the need to reduce time and cost.

The Aircraft Noise and Performance (ANP) database contains the type, noise, and performance characteristics of most aircraft. [8]

Usually aircraft types in ANP databases are grouped according to characteristic parameters like sound emission or/and the performance of the aircraft. Parameters which can be used to create categories might be: Maximum takeoff mass, type of engine, number of engines, by-pass ratio, etc.

In practice these parameters cannot be used separately, only combinations make sense. For instance, it is not useful to generate classes for aircraft with the same number of engines as engine types differ in noise. For this reason several parameters are needed to create relevant aircraft noise classes.

# 3 Aircraft Noise Models

Noise models, especially aircraft noise models, are needed for the prediction of noise levels where measurements are not feasible or possible. In most cases it is prohibitively expensive for airports to measure aircraft noise in their wider surroundings. For this reason noise models are commonly used to create mathematical representations of current noise levels or to predict future ones.

Typically, noise models compute sound exposure levels. There are several different approaches of calculation methods. Historically, almost every country used its own models and approaches. Nevertheless most aircraft noise models have in common that they are based on the mean value of the sound power of several flight events, where various times of the day (day, evening and night) are weighted differently.

The harmonization of aircraft noise figures is a concern of the European Commission. The acoustical basis measurement is proposed to be the mean-level $L_{Aeq}$. However, as aircraft noise can occur at different times of a day, the day-evening-night sound level $L_{den}$ is considered, too. Aircraft noise is quantified in Decibel, which is the ratio between the sound pressure of the aircraft and a reference pressure (the threshold of human hearing).

### *A-Weighting*

Depending on the frequency, the sound pressure is perceived differently at the human's ear. Several methods describe frequency-dependent weighting of noise. For aircraft noise models A-weighting [9] is most important, due to the fact that the A-weighting characteristic correlates quite well with people's subjective impressions concerning noise. The B-scale is an old and no longer used frequency weighting filter used to simulate a nominal human ear at about 70 dB. The C-weighting is used for peak sound levels near 100 dB or higher and due to its flat characteristic it is used to measure acoustic emission of machinery. The D-weighting, developed for loud aircraft noise measurements, is not widely used since the trend goes to the exclusive usage of the A-weighting filter. [10]

The "A"-filter approximates the sensitivity of the human's ear and helps to assess the relative loudness of noise. The curve of the described filter is shown in Figure 1. The X-Axis shows the frequencies in Hertz and the Y-Axis shows the A-weighted Decibel (dBA). It can be seen that the A-curve cuts off the lower and higher frequencies which an average person cannot hear. The A-weighting represents a constant loudness level of approximately 20 to 40 phon. [10]

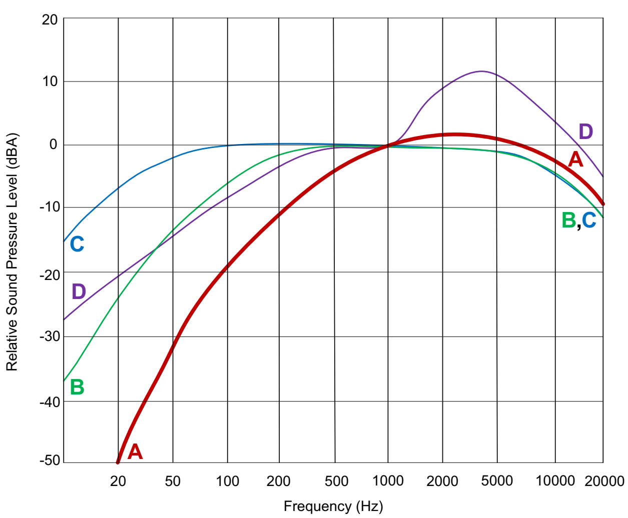



**Figure 1. Frequency Weighting**

### *Noise Level Indices*

In the following the primarily used noise level indices in this paper are explained. These are used for the description and calculation of aircraft noise models in this paper.

#### Sound Pressure Level $L_A(t)$

The time variable sound pressure level L(t) is a logarithmic dimension for the actual noise value at the location of a noise receiver. Via spectral filters (A-filter) the L(t) is adjusted for the frequency-dependent sensitivity of the human ear. This is denoted as $L_A(t)$ and can be seen in Figure 2.

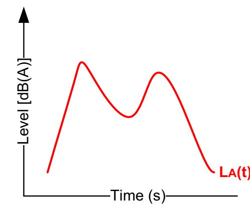



**Figure 2. Level-Time-Progress: $L_A(t)$**

#### Maximum A-Weighted Sound Pressure Level $L_{Amax}$

$L_{Amax}$ is the highest A-weighted noise level recorded during a noise measurement period from time $t_1$ to $t_2$ (see Figure 3 ). Metrics based on $L_{Amax}$ are mostly used in non-scientific settings. They are easier to measure and often much simpler to understand for the public, but do not take into account the duration of the noise. [11]

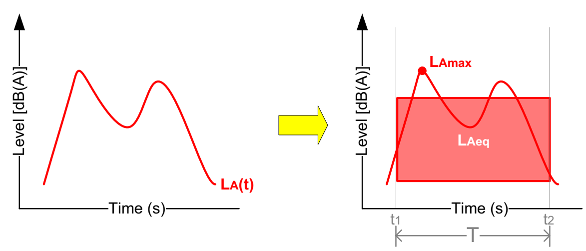



**Figure 3. Equivalent Continuous A-Weighted Sound Pressure Level $L_{Aeq}$**

#### Equivalent Continuous A-Weighted Sound Pressure Level $L_{Aeq}$

The equivalent continuous A-weighted sound pressure level $L_{Aeq}$ is widely used in the scientific literature as an index for noise. The $L_{Aeq}$ (see Figure 3) is calculated over a period of time T, and defined as the amount of average energy.

This value is often computed directly by noise models by using the aircraft-noise-model specific calculation of $L_A(t)$:

$$L_{Aeq} = 10 \cdot \log\left(\frac{1}{T}\int_{t_1}^{t_2} 10^{\frac{L_A(t)}{10}}\,dt\right)$$

### *Integrated Noise Model INM*

The **I**ntegrated **N**oise **M**odel INM [3] is a software product which is an implementation based on the Aircraft Noise Guidance ECAC Doc.29 3rd edition. The INM is the FAA's (Federal Aviation Administration) standard methodology for noise assessments since 1978. It is used in over 1000 organizations in over 65 countries. INM enables direct usage to assess noise impact with different metrics for various scenarios. For example:

- Runway configurations or new/extended runways
- New traffic demand and fleet mix
- Revised routings and local airspace structures
- Alternative flight profiles
- Modifications to other operational procedures

INM uses predefined approach and departure profiles from its database following a prescribed approach and departure procedure in terms of thrust and flap management.

INM provides three departure procedures (Standard, ICAO-A and ICAO-B [3]) for each aircraft type. It is up to the user to pick the departure procedure. As an additional feature, the usage of different take-off weights can change the altitude profile of an aircraft. But due to the fact that the aircraft weights are unknown in the radar data or simulations, various weight profiles cannot be used because of the loss of information. Therefore, a fixed profile is used for the calculations.

Unlike supporting three different departure procedures, INM supports only one approach procedure for each aircraft type. Each type is associated with exactly one approach profile. Most of the airplanes in INM use a 3-degree descent angle, better known as Continuous Descent Approach (CDA) to model IFR flights. For single engine airplanes a 5-degree descent angle is used to model VFR flights. Nevertheless INM also uses the so-called "Standard approach" for some aircrafts.

As a matter of fact the 3-degree descent angle cannot be used during peak times. The intense air traffic during the day makes it almost impossible to use the approach procedure on larger airports. The continuous approach is too often aborted due to crossing air traffic or due to the maintenance of the vertical separation. During the night the air traffic declines and therefore the continuous descent approach can be assigned by the air traffic control more often. [12]

The predefined approach profiles for each aircraft, which are used for the aircraft noise calculation with INM, differ therefore with the reality.

### *FLULA2 Noise model Method*

FLULA2 [6] is a noise calculation method, which has been developed by the Swiss company EMPA, and is used for the calculation of aircraft noise exposure on military and civil airports. The calculations are based on aircraft noise measurements for starting and landing aircraft at the airport Zurich/Kloten (LSZH) of the year 1996. The results of these measurements were 32 coefficients, which implicitly include the frequency dependence of air damping for atmospheric standard conditions. The coefficients of the most common different aircraft types are used to describe the directional characteristics of the mathematical model. It was deemed necessary for the correct calculation of spiraling which is not guaranteed with other noise models.

Besides the direction-dependent sound emission, the air absorption and the distance attenuation are packed into the mathematical model. In the simulation this is used to get a noise value for every receiving point and furthermore to reconstruct the sound level course.

All necessary values, sound level $L_{Amax}$ and event level $L_{AE}$ for the evaluation of aircraft noise calculation are the result of the chronological sequence of the noise level. The summation of the events of all single flights produces the overall impact, the equivalent continuous sound level $L_{Aeq}$.

Through the use of radar data, single flight movements can be considered. But in the same way these great amount of data needs much calculation time. Big differences to other models, which use international available data bases, are the locally needed noise measurements for the mathematical model. Meteorological effects like wind and weather are not considered.

### *AZB Noise Model Method*

The AzB [7] Noise Model Method in its last version of the year 2008 is based on the "Law for protection against aircraft noise". The AzB2008 method is a segmentation model and was developed to calculate aircraft noise for long-term forecasts – the six busiest month of the forecast year are the base for the noise calculation.

For the evaluation and comparison of different aircraft noise models methods it was necessary to show a single flight and its noise contour and single noise values over time.

The assumption underlying the AzB model is a moving point-acoustic-source, for which the sound exposure at every point of the flight path, the speed, and the radiation characteristics are known. The moving point-acoustic-source is reproduced with a line-acoustic-source which is used for the emission calculation. These moving acoustic sources represent the aircraft classes of AzB.

This noise model is based on different groups of aircraft which are used for the calculation. Each group has at least one class for takeoff and another one for landing. The classes are used to describe the acoustical and operating flight characteristics.

## 4 Common Database Interface

Due to different noise model calculation approaches, various input data sets (radar data, simulated data, etc.) are needed. To achieve a more efficient computation of different noise models on the basis of unified input data formats, the author designed an uniform interface, which allows to integrate data sets of different sources with all noise models. For the software tool INM a special adapter was designed to produce the exact INM data format. Figure 4 shows the design for aircraft noise calculation.

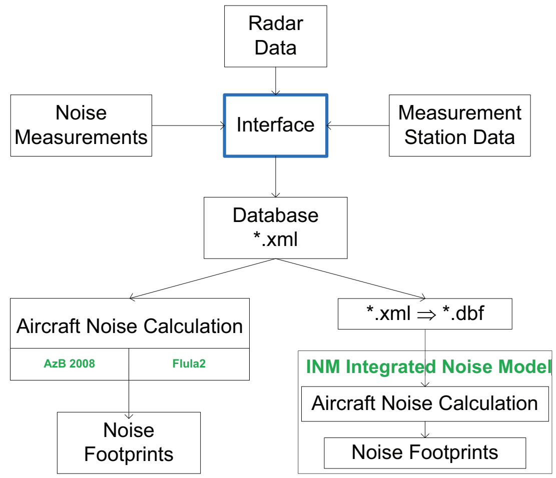



**Figure 4. Common Database Interface**

### *Radar Data and Noise Measurements*

The German Aircraft Noise Service ("Deutscher Fluglärmdienst", short DFLD) is a non-profit organization, founded in 2002, which is dedicated to collect all air traffic emissions and their transparent illustration as well as long-term archives.

With almost 400 measurement stations operating in over 40 regions and 7 countries (Germany, France, Greece, Great Britain, The Netherlands, Austria and Switzerland) the DFLD has an enormous amount of data. With maximal one hour delay these data is published on their website [13].

Flight Tracks are available for download in compressed kml-file-format (Keyhole Markup Language), which is a markup language for the description of geographical data for Google Earth and Google Maps. KML follows the XML-Syntax and therefore it was possible to parse these files and extract the needed information for aircraft noise calculation into a compressed predefined XML-format which is used as input for the aircraft noise calculation methods FLULA2, AzB, and the software tool INM.

For each measurement station, published on the DFLD Website, the noise measurements are shown in a time-dependent profile. These real noise measurement data is used to validate the implemented noise model methods FLULA2 and AzB.

### *Virtual Noise Measurement Stations*

Virtual noise measurement stations are generated to simulate a grid of location points on the ground in the vicinity of an airport. The grid of virtual stations is needed to compute an aircraft noise contour. With latitude and longitude of the generated grid point, the exact altitude of each point is calculated by using the SRTM-data (Shuttle Radar Topography Mission) [14].

### *Noise Model Input*

For the noise calculation with various noise model approaches it is irrelevant which dataset was originally used to generate the database. Every noise model approach, that the author implemented, can use the same input format for the respective computations. For INM, the author also used this input to generate new files with a very special file structure and file type (`DBASE` files) required by INM.

## 5 Validation and Results

### *Aircraft Noise Contours*

For the validation of aircraft noise contours, the equivalent continuous sound level $L_{Aeq}$ was calculated. The following images show noise contours of the noise calculation approach with INM 7.0, FLULA2, and AzB2008.

**Departure**

Figure 5, Figure 7 and Figure 9 show the departure (red line) on Salzburg Airport (LOWS) with an Airbus A320.

It can be seen that AzB2008 doesn't perform as well as the other models at the beginning of the departure. INM 7.0 and FLULA2 simulate a more accurate departure with a two-engine aircraft as it can be seen in the convexities on the lower end of the runway. AzB2008 is a noise model method developed for long-term periods instead of single flight evaluations which could explain the different behavior at the beginning of the departure.

Nevertheless these contours show that the color ranges directly below the flight path are very similar.

**Arrival**

The arrival contours (Figure 6, Figure 8 and Figure 10) are not as similar as the departure contours. INM 7.0 has the smallest contour, while FLULA2 and AzB2008 has wider color areas below the flight path. A big difference between INM 7.0 and the other models is the approach procedure. INM 7.0 uses a standard arrival procedure predefined in the software, which affects the altitude profile of the approach. For the Airbus A320 the standard procedure in INM 7.0 is the Continuous Descent Approach (CDA). FLULA2 and AzB2008 use the altitudes given in the radar data.

It can be clearly seen that INM 7.0 and AzB2008 accounts reverse thrust after touchdown which widens the contour when the aircraft starts the deceleration process on the runway. FLULA2 on the other side does not account the reverse thrust after touchdown and has therefore a very different noise contour in this approach phase.

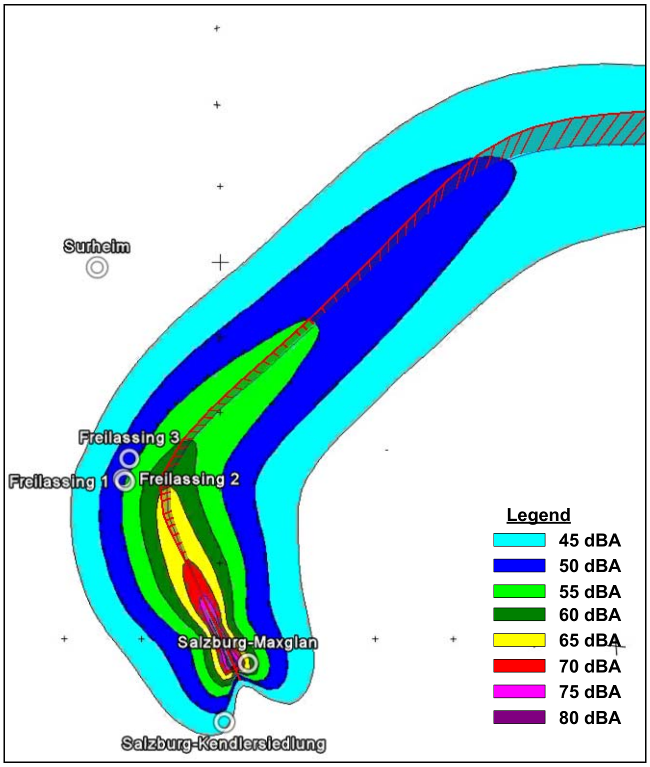



**Figure 5. Departure Calculation with INM 7.0**

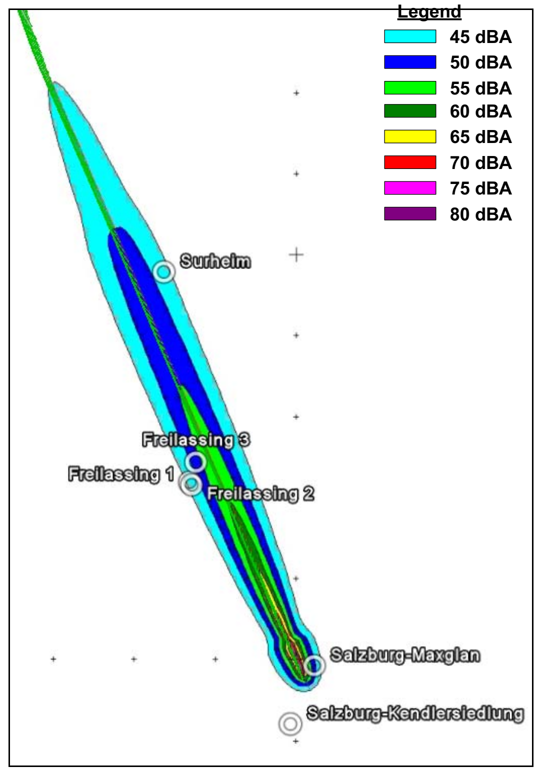



**Figure 6. Arrival Calculation with INM 7.0**

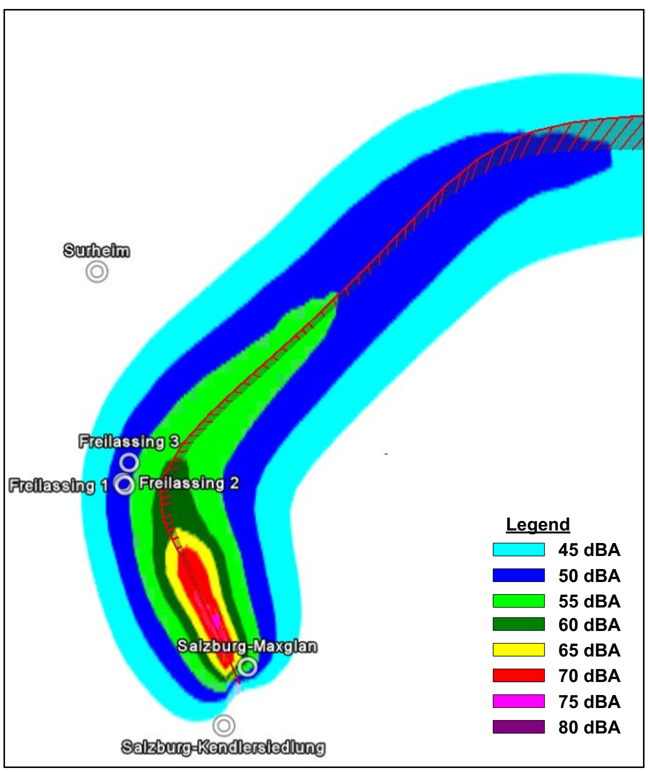



**Figure 7. Departure Calculation with FLULA2**

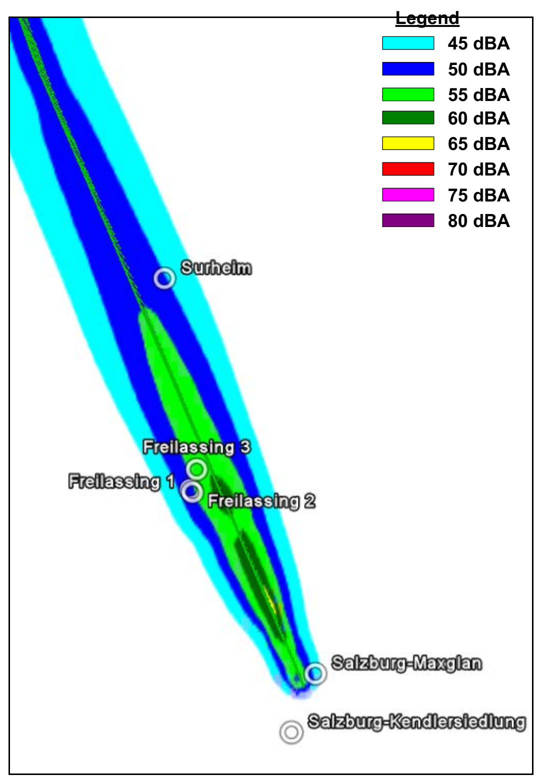



**Figure 8. Arrival Calculation with FLULA2**

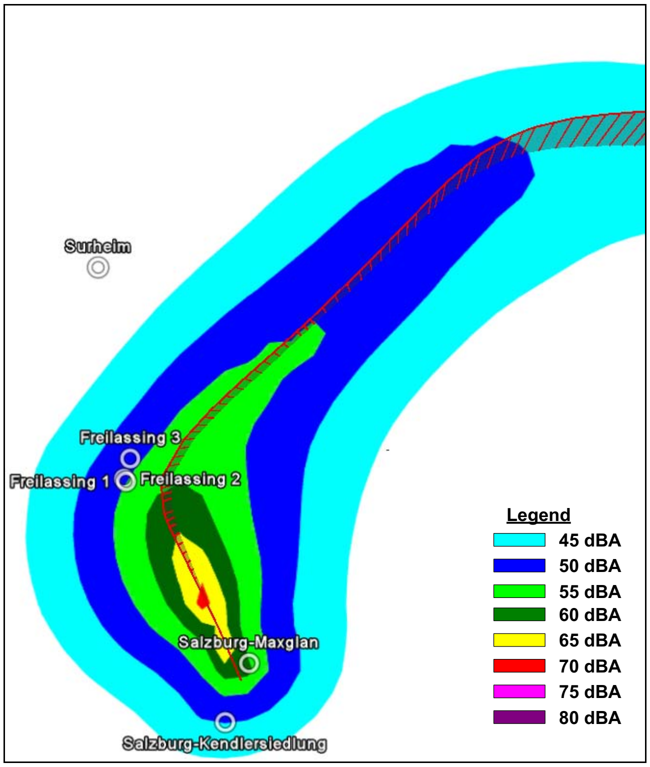



**Figure 9. Departure Calculation with AzB2008**

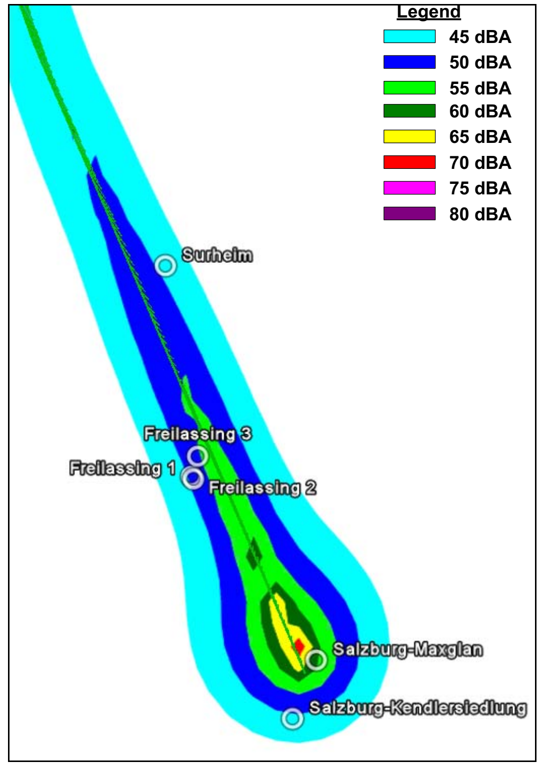



**Figure 10. Arrival Calculation with AzB2008**

### *Real Measurement Data*

Due to the fact that real noise measurements are measured on only a few points in the vicinity of an airport, the generation of noise footprints from measurement data is not feasible. However, it is possible to compare the measured noise values with the generated values of noise models for the available measurements.

### *Aircraft Noise Measurement Stations*

In Figure 11 the arrival (green) and the departure (red) on Salzburg airport are illustrated in Google Earth. The figure also shows the six measurement stations, which are provided by the DFLD [13]. Their position is used for the noise measurement comparison between calculated noise model approaches and the measured noise values.

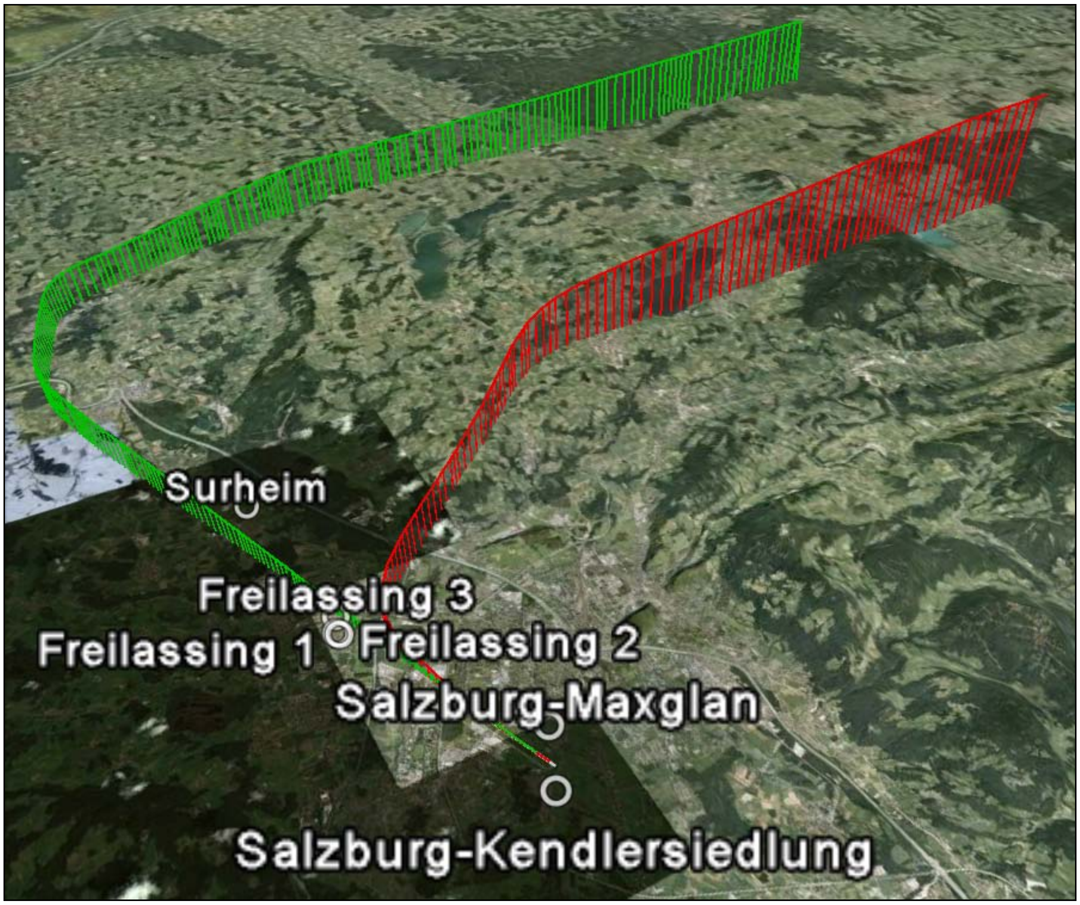



**Figure 11. Noise Measurement Stations**

## *Results*

For the validation of aircraft noise models with real measurement data, the single noise values $L_A(t)$ had to be calculated.

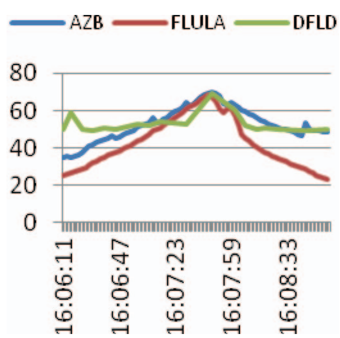
To validate the aircraft noise calculation models FLULA2 and AzB2008, the author collected the data on the DFLD website for comparison. INM 7.0 does not provide the single noise results $L_A(t)$ and therefore cannot be used for comparison.

Other sources of error can occur through wind, rain, reflection and other disturbing sources around the noise measurement station. These facts have to be considered reading the following comparisons between calculated and measured noise data.

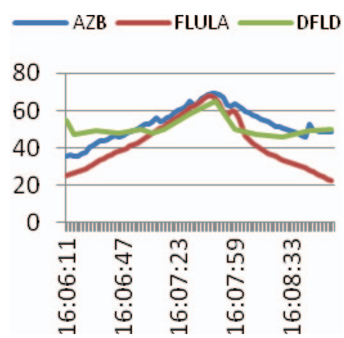
### *Arrival*

The results of the aircraft noise calculation approach models FLULA2 and AzB2008 and the real measurements collected by the DFLD are illustrated in Table 1. It can be seen that AzB2008 (blue graph) and FLULA2 (red graph) deliver very similar noise values. In the first four graphics the peaks ($L_{Amax}$) of calculated and measured data are almost identical.

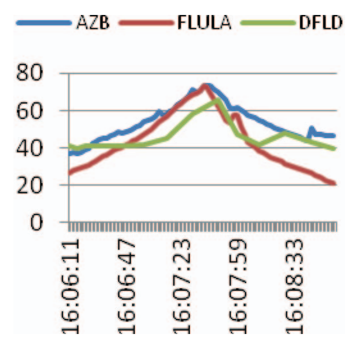
**Table 1. Approach Table Comparing Real Measurements vs. Calculation**



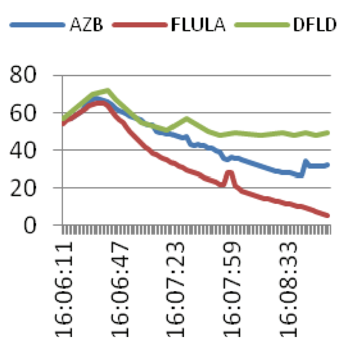

**Freilassing 1**



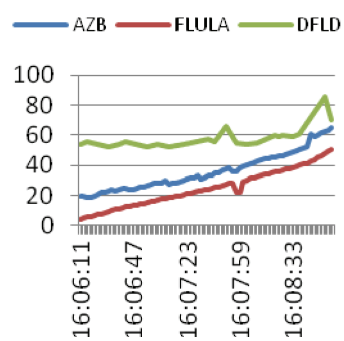

**Freilassing 2**



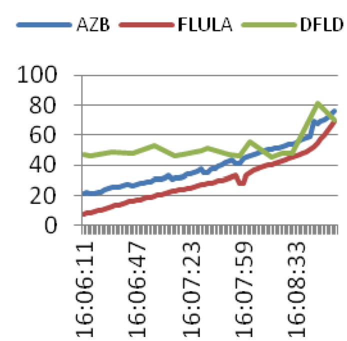

**Freilassing 3**




**Surheim**




**Kendlersiedlung**



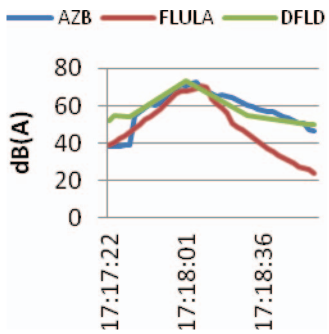

**Maxglan**

The noise values given in the graphs are the chronological sequence of the flight. The used noise metric therefore is $L_A(t)$ as described earlier in this paper. Only the last part of the flight these noise calculation approaches have more deviant values. AzB2008 shows in the end slightly louder noise values than FLULA2, which was already discussed before, where the noise contours showed this phenomenon. Compared to the real measured data, AzB2008 supplies more accurate data in the last phase of the arrival. A reason that FLULA2 is not as intense as AzB2008 could be, that the 32 coefficients needed for the calculation are measured around airport Zurich Kloten and not Salzburg airport. This is also the biggest disadvantage of FLULA2. Every airport would need the predefined coefficients, received from the complex measurements in the vicinity of the airport, to get appropriate noise values.

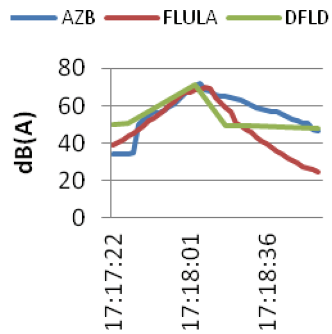
### *Departure*

Table 2 illustrates the graphics of a departing aircraft Airbus A320 on Salzburg Airport. The results are similar to the approach. Only at the first phase of the departure at the airport runway the noise values vary (Measurement stations: Kendlersiedlung and Maxglan). The other stations have almost identical noise values at their peaks ($L_{Amax}$).

**Table 2. Departure Table Comparing Real Measurements vs. Calculation**




**Freilassing 1**

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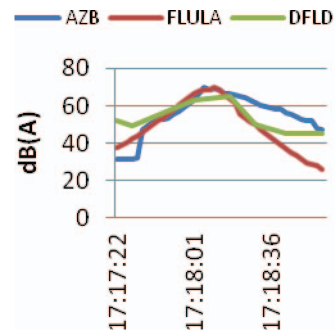


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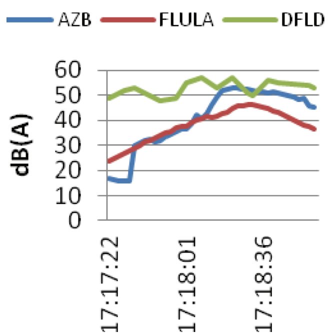


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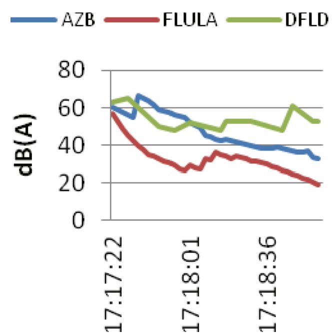
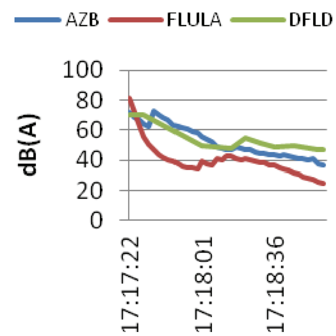



**Kendlersiedlung**




**Maxglan**

## 6 Conclusion and Outlook

In this paper the author described several aircraft noise calculation methods, determined their effectiveness, discussed the resulting aircraft noise contours, and validated the good correlation of real noise measurements and calculated noise values. To achieve a good overview of the effectiveness of the described aircraft noise methods the author chose to use a single selected flight. This is the best approach to investigate the differences of the inspected methods.

As discussed in this paper, calibrated noise measurement stations would be needed to eliminate such systematic errors. Beyond that a good chronologic sequence of noise measurements is needed to achieve a good quality for comparisons between calculation models and measurements.

Another interesting point for further research are noise abatement measures, e.g. night flight restrictions, noise contingents, noise reduction procedures, etc. As for restrictions and noise contingents it is necessary to group airplanes into aircraft noise classes, which can then be used as hypotheses in noise abatement measures. For example a louder noise class is no longer allowed to land on an airport during the night which affects the noise contours. These measures are targeted on airlines to refit their airplanes (for example with new and quieter engines). A refitted airplane is then shifted to another noise class which satisfies the restrictions.

Other measures are targeted on reducing aircraft noise during approach or departure procedures. One of these measures is already "predefined" in INM 7.0. The approach profiles of most aircraft types are using the Continuous Descent Approach (CDA), which is not realistic as discussed in this paper. The aircraft noise contour received with standard settings in INM 7.0 results in a noise contour which already includes noise abatement procedures. For more realistic approach flight profiles in INM 7.0 new flight profiles for each aircraft need to be created depending on the airport. More frequented airports are not able to use the CDA during day times as discussed in this paper.

Due to complexity some noise influencing factors are still not considered in aircraft noise methods. For example wind and weather is not considered in any of the regarded aircraft noise models in this thesis due to the complex mathematical methods.

## References

[1] Maute, Dieter, 2006, Leipzig, Carl Hanser Verlag.

[2] Bundesminister des Inneren, 1975, Anleitung Zur Berechnung Von Lärmschutzbereichen An Zivilen Und Militärischen Flugplätzen Nach Dem Gesetz Zum Schutz Gegen Fluglärm Vom 30. Maerz 1971.

[3] Eric R. Boeker et al., 2008, Integrated Noise Model (INM) Version 7.0 Technical Manual, Washington, DC, Federal Aviation Administration, Office of Environment and Energy.

[4] European Civil Aviation Conference (ECAC), 2005, Report on Standard Method of Computing Noise Contours around Civil Airports, Volume 1, Applications Guide, France.

[5] European Civil Aviation Conference (ECAC), 2005, Report on Standard Method of Computing Noise Contours around Civil Airports, Volume 2, Technical Guide, France.

[6] Thomann, G., Bütikofer, R., Krebs, W., 2005, FLULA2 – Ein Verfahren zur Berechnung und Darstellung von Fluglärmbelastung. Technische Programmdokumentation, Version 2.1, Dübendorf (CH).

[7] Bundesminister des Inneren, 2008, Anleitung zur Berechnung von Lärmschutzbereichen an zivilen und militärischen Flugplätzen nach dem Gesetz gegen Fluglärm vom 31. Oktober 2007.

[9] Horonjeff, R., McKelvey, F.X., Sproule, W.J., Young, S.B., 2010, Planning and Design of Airports, McGraw-Hill Professional, U.S.A.

[10] Crocker, M.J., 1998, Handbook of Acoustics, John Wiley & Sons, New York, U.S.A.

[11] Mensen, H., 2004, Moderne Flugsicherung, Springer Verlag, Berlin – Heidelberg.

[8] Eurocontrol, The Aircraft Noise and Performance (ANP) database, http://www.aircraftnoisemodel.org

[12] DFS Deutsche Flugsicherung GmbH, 2008, CDA – Continuous Descent Approach, Langen

[13] Deutscher Fluglärmdienst e.V., http://www.dfld.de

[14] NASA: Jet Propulsion Laboratory, Shuttle Radar Topography Mission, http://www2.jpl.nasa.gov/srtm/

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